

Yongqi Bai

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RESEARCH INTERESTS

Reinforcement learning and world-model planning for LLM/VLM agents; multi-turn decision-making under partial observability; applications to quantitative finance

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

Hong Kong SAR

B.Sc. in Mathematics & Quantitative Social Analysis (Double Major)

Sep 2024 – Jun 2028

Relevant Coursework: Probability, Statistics, Linear Algebra, Multivariable Calculus

University of Virginia (UVA) | Semester Exchange

Fall 2026

Coursework (Fall 2026): Basic Real Analysis, Financial Time Series & Forecasting, Regression Analysis, Foundations of Machine Learning, Theory of Financial Markets, International Finance & Macroeconomics

RESEARCH EXPERIENCE

Research Assistant, Dept. of Computer Science & Engineering, HKUST

Feb 2026 – Present

Advisors: Prof. Song Guo (Chair Professor) & Prof. Jie Zhang

- **Co-author on a manuscript in preparation for a top-tier AI conference** — extending adaptive-lookahead world-model planning (“Imagine-then-Plan”) from text-only LLM agents to vision-language agents (POIMDP formulation; VAGEN/VerL stack)
- **Diagnosed that adaptive lookahead pays off only on an RL-trained base policy** via a 150-scenario (50-seed $\times$ 3-mode) zero-shot ablation ($K = 0/1/\text{adaptive} \rightarrow 24\%/16\%/20\%$) — the finding re-scoped the group’s research roadmap
- **Lifted validation success 27.3% $\rightarrow$ 37.9%** in a 150-step distributed GRPO run (Qwen2.5-VL-3B; 4 $\times$ NVIDIA H800; FSDP + async sglang rollouts on Slurm) — train reward $\sim 4\times$, actor entropy 1.04 $\rightarrow$ 0.29, zero failed runs
- **Built a 53,567-sample expert dataset** from 3,000 self-collected ALFWorld trajectories (95% expert success), in SFT and next-state world-model formats
- **Resolved a deep ABI conflict on a read-only shared cluster** (verl 0.6.1 $\times$ sgl_kernel) via four targeted import guards and PYTHONPATH shadowing — zero modifications to the lab-shared environment

Systematic Factor Investing & Constrained Portfolio Optimization (*Independent Research*)

2024 – Present

Python (pandas, NumPy, cvxpy) $\cdot$ S&P 500 universe (~ 500 names) $\cdot$ monthly dollar-neutral long/short, 2016–2026

- Built an **end-to-end factor research pipeline**: 24 factors (15 price/volume + 9 point-in-time SEC EDGAR fundamentals), sector/beta neutralization with IC admission gates, Ledoit-Wolf covariance, and a fully-constrained cvxpy mean-variance optimizer (neutrality, leverage, per-name and turnover caps)
- **Reported a rigorous negative result**: nothing cleared Deflated Sharpe 0.95 across an $N = 38$ trial registry, 28-path Combinatorial Purged CV, and a pre-registered one-shot holdout — the verifiable ceiling on free data is ≈ 0.66 gross (0.58 / 0.50 net at 10 / 20 bp)
- **Sharpe 0.47 $\rightarrow$ 0.67 from portfolio construction alone** — same neutralized momentum signal, equal-weight quintile long/short vs. the fully-constrained optimizer
- **Applied the framework to fund attribution** (*companion study*) — decomposed multi-cap quant funds’ excess returns from NAV series alone via BARRA-style return-based regression, separating genuine selection alpha from style (size/value/momentum) exposure

INDUSTRY EXPERIENCE

Huatai United Securities — Investment Banking Division, Intern (Shanghai)

May 2026 – Present

- **Independently authored a 41-page industry study** on optical interconnects / CPO with a 10-company deep-dive, anchored in first-hand teardowns of US value-chain leaders and held to prospectus and due-diligence-memo standards
- **Drafted the equity-story (OSS) section of an in-progress HKEX Main Board IPO prospectus**, synthesizing 11 filings and diligence sources into a three-pillar thesis — first draft adopted as the team’s working base

CASH Financial Services Group — Securities Trading Intern, Wealth Management (Hong Kong)

Feb 2026 – Apr 2026

- **Issued a BUY initiation on Laopu Gold (6181.HK)** with an explicit cost-of-capital stack (Ke 11.7%, WACC 9.4%, terminal 3.0%) and base/bull/bear DCF scenarios to an HKD 1,296 target, benchmarked against Chow Tai Fook
- **Researched cross-asset trade ideas with scenario-stressed P&L and explicit sizing** for the desk PM — rates relative value (2s10s steepener; DV01-neutral UK–US 2Y widener), FX and options expressions

CITIC Securities — Investment Advisory Intern (Xi’an)

Dec 2025 – Feb 2026

- **Modeled OTC structured-product payoffs in Python** (barrier notes, index-enhanced structures), mapping exposure across strikes and barrier levels to support client derivatives discussions

SKILLS & ADDITIONAL

Technical: Python (PyTorch, transformers, pandas, NumPy, cvxpy), distributed training (FSDP, DeepSpeed, Slurm/HPC), R, SQL, C++, LaTeX, Bloomberg Terminal, Wind

Languages: Mandarin (Native), English (Fluent)

Interests: Basketball (varsity point guard, 3 $\times$ league champion), Marathon (2 $\times$ finisher), Golf